

Evidence Summary - Technology #08

NIR Photovoltaic Power Delivery for Implantable Devices

External Evidence Sources (5 total):

Source 1: Subcutaneous Photovoltaic Infrared Energy Harvesting for Bio - PMC

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5646820>

Excerpt: The PV cells may be stacked on the implantable system, and include an integrated photodiode to provide a means of low-power wireless optical communication to ...

Source 2: [PDF] Subcutaneous Photovoltaic Infrared Energy Harvesting for Bio

URL: <https://www.semanticscholar.org/paper/Subcutaneous-Photovoltaic-Infrared-Energy-for-Moon-Blaauw/18fae96f6f7351833fa96dd7f5e22af54375394d>

Excerpt: This work shows that photovoltaic cells at the millimeter-scale can achieve a power conversion efficiency of more than 17% for silicon and 31% for GaAs ...

Source 3: GaAs-Based Photovoltaic Infrared Energy Harvesting for Microscale

URL: https://www.researchgate.net/publication/365383611_GaAs-Based_Photovoltaic_Infrared_Energy_Harvesting_for_Microscale_Biomedical_Implants

Excerpt: GaAs-Based Photovoltaic Infrared Energy. A method for efficient, rapid, and minimally invasive implantation ■ utilizes custom-designed PV cells specifically ...

Source 4: Near-Infrared Subcutaneous Generator for Powering Implantable

URL: <https://journal.sitp.ac.cn/hw/hwen/article/html/H2026025>

Excerpt: This article reviews the latest advances in light transmission enhancement strategies and energy conversion mechanisms in near-infrared subcutaneous generators.

Source 5: [PDF] Dual-Junction GaAs Photovoltaics for Low Irradiance Wireless Power

URL: <https://blaauw.engin.umich.edu/wp-content/uploads/sites/342/2021/02/Dual-Junction-GaAs-Photovoltaics-for-Low-Irradiance-Wireless-Power-Transfer-in-Submillimeter-Scale-Sensor-Nodes.pdf>

Excerpt: Feb 3, 2021 · Near-infrared (NIR) light provides a means for high-power conversion efficiency (>30%) in millimeter-scale subcutaneous implantable devices [14] ...

Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions. No evidence has been physically validated.